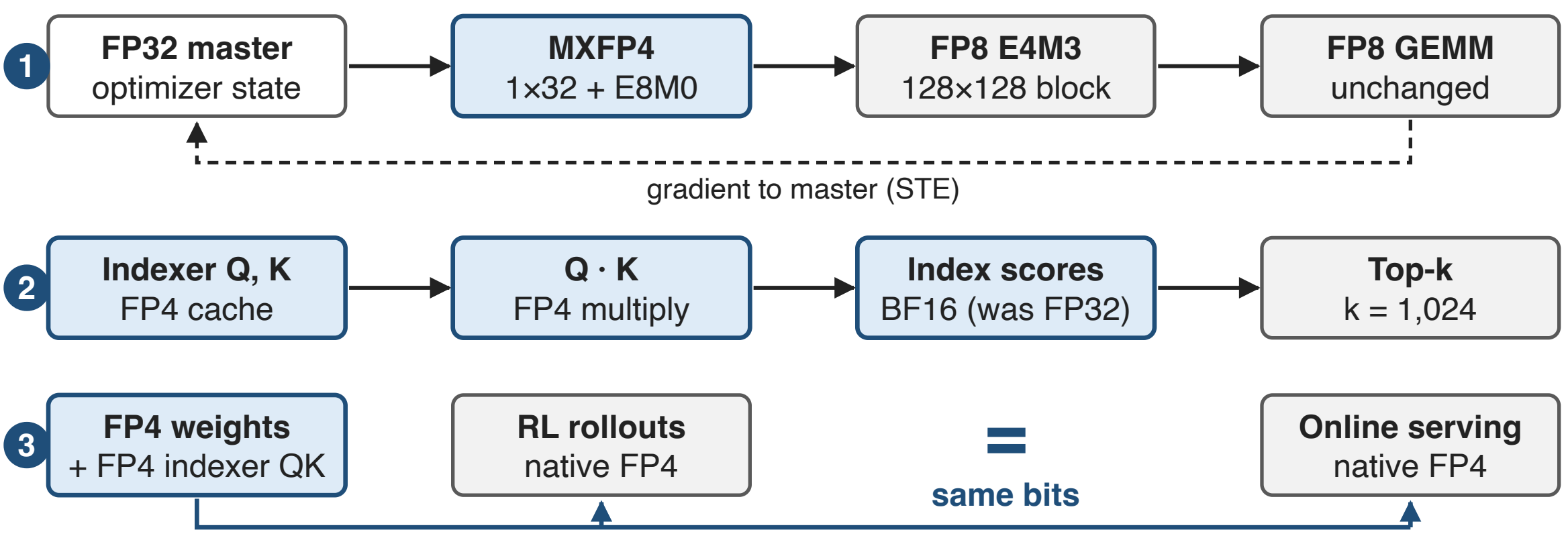
1
FP32 master
optimizer state
MXFP4
1×32 + E8M0
FP8 E4M3
128×128 block
FP8 GEMM
unchanged
gradient to master (STE)
2
Indexer Q, K
FP4 cache
Q · K
FP4 multiply
Index scores
BF16 (was FP32)
Top-k
k = 1,024
3
FP4 weights
+ FP4 indexer QK
RL rollouts
native FP4
=
same bits
Online serving
native FP4